ITC 508 (Deep Learning and NLP)

Final Project: RAG Chatbot Development and Evaluation

Note: These are the deliverables for each group for the final project. The files will be used to guarantee that the claims made in the final IEEE paper's Methodology and Results are auditable and reproducible and that the workload was appropriately distributed to all group members and all members participated in the technical execution. A non–compliance to any requirements will deem the final submission as void.

Part I. For Ensuring Scientific Verification (Paper Content is True & Correct)

Deliverable	Purpose and Verification Check	File Type(s) Required
1. **Final Draft of IEEE Report (Authorship Included)**	The comprehensive IEEE report detailing the experiments' details, methodology and results. Must include all authors' names.	.docx
2. **Chunking Experiments**	Documents at least three chunk-size and overlap configurations, actual experiment outputs, comparisons, and justification for the selected configuration.	.ipynb, .xlsx
3. **Source Code and Reproducibility Package**	Contains the complete RAG implementation, chatbot application, and experimentation scripts.	.ipynb and supporting files
4. **Experiment Logs and Checkpoints**	Provides actual execution outputs from document processing, embedding generation, vector database construction, retrieval, LLM integration, and chatbot evaluation.	.xlsx and supporting files
5. **Chatbot Response Evaluation**	Contains at least 10 evaluation questions with corresponding reference answers, actual chatbot-generated responses, and comparative assessment results. Both answerable and unanswerable questions must be represented.	.csv, .xlsx

Part II. For Ensuring Individual Contribution (Accountability)

Deliverable	Purpose and Verification Check	File Type(s) Required
6. **Member Responsibility Matrix (MRM)**	A formal table detailing each member's specific, non-overlapping primary tasks.	.docx

Deliverable	Purpose and Verification Check	File Type(s) Required
7. Signed Final Project Integrity and AI Usage Statement	Document confirming the authenticity of the submitted work, accuracy of reported results, individual contributions, and proper disclosure of generative AI usage.genuine student efforts and not through the use of generative AI.	.pdf

General Submission Requirements

1. All seven deliverables must be submitted in their prescribed formats.
2. All source code, experiment results, screenshots, and evaluation outputs must correspond to the actual chatbot implementation.
3. The chatbot must demonstrate the integration of document processing, chunking, embeddings, vector storage, retrieval, and LLM-based response generation.
4. Each group must provide all the project source code and supporting documentation.
5. Each member must provide identifiable evidence of their technical contributions and be prepared to explain and defend their assigned component.
6. All groups must ensure that their evaluation test set contains at least 10 questions, with corresponding reference answers, including both answerable and unanswerable questions.
7. All members must comply with the course's academic integrity and AI usage requirements.

IMPORTANT: Non-compliance with any of the required deliverables may render the final project submission void.